Irem **Topal**

Physics Engineer | PhD Candidate

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Education

2019 - present	PhD in Computational Science and Engineering, Kadir Has University, Istanbul, Turkey.
2019	PhD in Physics Engineering, Istanbul Technical University, Courses Transferred.
2017	MSc in Physics Engineering, Istanbul Technical University.
2014	BSc in Physics Engineering, Istanbul Technical University.

Research and Experience

Present – October 2019 | **Research Assistant, Kadir Has University, Istanbul, Turkey.**
PhD funded by TUBITAK

Sept 2021 – July 2021 | **Visiting PhD Student, International Center for Theoretical Physics, Trieste, Italy.**
Quantitative Life Sciences Section

October 2019 – November 2018 | **Physics Tutor, Bahcesehir, Istanbul, Turkey.**
Worked one-on-one with undergraduate engineering students to prepare them for the exams in Physics I and II.

October 2019 – April 2018 | **Data Scientist, Miletos Inc., Istanbul, Turkey.**

Sept 2013 – August 2013 | **Engineering Intern, Eczacibasi-Monrol Nuclear Products, Kocaeli, Turkey.**

August 2013 – July 2013 | **Intern, Istanbul University, Cerrahpasa Faculty of Medicine, Istanbul, Turkey.**
Received Nuclear Medicine Physics Certificate.

Publications

2023 Topal, I., Belousov, R. and *et al.*"*Machine-learning classification of blood-flow malformations in cerebral vessels*" (*Manuscript in preparation.*)

2023 Topal, I., Eroglu, D., *"Reconstructing Network Dynamics of Coupled Discrete Chaotic Units from Data"*, Physical Review Letters **130**, 117401, (2023).

2019 Topal, I., Servantie, J., *"Molecular Dynamics Study of the Thermal Conductivity in Nanofluids"*, Chemical Physics **516**, 147-151, (2019).

Talks and Posters

2022 *Reconstructing network dynamics of weakly coupled chaotic units from data* (Talk), Inverse Network Dynamics, Max Planck Institute for the Physics of Complex Systems, Dresden, Germany.
2022 *Caliban and the Witch : Professions Lost by Women* (Podcast), Fiskos Masası, Kadir Has University, Istanbul.
2021 *Introduction to Network Sciences* (Online Lecture), 19th Physics Days, Istanbul.
2021 *Network Reconstruction from Data* (Poster), The EPS conference on Statistical Physics of Complex Systems, ICTP, Trieste.
2021 *Reconstructing Network Dynamics of Weakly Coupled Chaotic Units* (Online), SIAM Conference on Applications of Dynamical Systems.
2021 *Synchronization of Chaotic Systems* (Online Lecture), 17th Physics Days, Istanbul.
2020 *Sensing Network Dynamics of Weakly Coupled Chaotic Units* (Online), Dynamics Days Digital 2020.
2018 *Prediction of DNA Binding Sites for a Transcription Factor* (Poster), Machine Learning Summer School at Bosphorus (BYOYO), Istanbul.
2018 *Molecular Dynamics Study of the Thermal Conductivity in Nanofluids* (Talk), The annual Statistical Physics Days, İzmir.

Skills

Programming	Mathematica, Maple, Julia
Software	VMD, Inkscape, LaTex, Grace, Gephi
Development tools	git
Operating systems	Ubuntu, Mac OS X, Windows

Programming Languages

Python	●●●●●
Fortran	●●●●○
MATLAB	●●●●○

Languages

- Turkish (Native)
- English

Outreach and Volunteering

2013 İstanbul Technical University, Students' Representative Council, Undergraduate Student Representative of Physics Engineering

References

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